

SUGGESTIONS FOR RARITONE

Introduction

During the internship, our team explored various AI-powered technologies related to virtual try-on systems, fashion recommendation engines, body measurement estimation, and AR/VR-based shopping experiences. Based on our research findings and observations, we have identified several opportunities that can help Raritone further enhance its platform and provide a more personalized, engaging, and efficient shopping experience for users.

1. Implement Real-Time Virtual Try-On

One of the most impactful improvements would be the development of a real-time virtual try-on feature. Instead of static image-based previews, users could instantly visualize clothing on themselves through live camera input. This would improve user confidence, reduce uncertainty during purchases, and create a more interactive shopping experience.

Benefits:

- Improved customer engagement
- Enhanced shopping experience
- Increased purchase confidence
- Reduced product return rates

2. Develop an AI Personal Stylist

An AI-powered personal stylist could provide intelligent outfit recommendations based on user preferences, body type, fashion trends, and purchase history. Such a feature would make fashion discovery easier and help users find products that match their individual style.

Benefits:

- Personalized shopping experience
- Better customer satisfaction
- Increased product discovery
- Higher user retention

3. Improve Body Measurement Estimation

Accurate body measurements play a critical role in virtual fitting systems. Future development could focus on improving measurement extraction from user images by combining pose estimation, body landmark detection, and advanced AI models.

Benefits:

- Better garment fitting accuracy
- More realistic virtual try-on results
- Improved size recommendations
- Enhanced user trust

4. Integrate Advanced 3D Virtual Try-On

While 2D virtual try-on systems are effective, 3D virtual try-on solutions provide a more immersive and realistic experience. Future versions of the platform could incorporate 3D avatars, body mesh reconstruction, and realistic cloth simulation to improve visualization.

Benefits:

- Realistic garment representation
- Better fit visualization
- Enhanced customer experience
- Competitive advantage in the fashion technology market

5. Introduce AR-Based Shopping Experiences

Augmented Reality (AR) technology can allow users to virtually wear clothing, accessories, footwear, and jewelry through their smartphones. AR integration would provide an engaging and modern shopping experience while reducing uncertainty before purchase.

Benefits:

- Interactive product visualization
- Improved customer engagement
- Increased conversion rates
- Innovative shopping experience

6. Expand Fashion Datasets

The performance of AI systems depends heavily on the quality and diversity of available data. Expanding datasets to include multiple body types, clothing styles, cultural fashion preferences, and pose variations can improve model accuracy and inclusiveness.

Benefits:

- Better model performance
- Improved recommendation quality
- Support for diverse user groups
- More accurate virtual try-on outputs

7. Enhance Recommendation Systems

Future recommendation engines can move beyond basic filtering techniques by incorporating user behavior analysis, seasonal trends, social media influences, and AI-generated fashion insights.

Benefits:

- More relevant recommendations
- Increased customer engagement
- Better shopping personalization
- Higher sales opportunities

8. Mobile Application Integration

As mobile shopping continues to grow, integrating AI-powered features directly into a mobile application can significantly improve accessibility and convenience for users. Features such

as virtual try-on, AI recommendations, and personalized styling can be made available on smartphones.

Benefits:

- Improved accessibility
- Greater user convenience
- Increased platform reach
- Enhanced user experience

9. Optimize AI Performance and Scalability

As the platform expands, optimizing AI workflows for speed and scalability will become increasingly important. Lightweight models, efficient processing pipelines, and cloud-based deployment strategies can help ensure smooth performance even with a large number of users.

Benefits:

- Faster response times
- Improved scalability
- Better user satisfaction
- Reduced computational costs

Conclusion

The research conducted during the internship demonstrates the significant potential of Artificial Intelligence, Computer Vision, and AR/VR technologies in transforming the online fashion industry. By implementing these recommendations, Raritone can further strengthen its position as an innovative fashion technology platform and provide users with highly personalized, interactive, and intelligent shopping experiences. These improvements can contribute to increased customer satisfaction, higher engagement levels, and long-term platform growth.